

V-band Time-Series Photometry of AM CVn: Independent Recovery of the Superhump and Its First Harmonic

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Abstract

AM Canum Venaticorum (AM CVn) is the prototype of a variable, hydrogen-deficient binary system in which a white dwarf accretes matter from its compact hydrogen-poor donor via an accretion disk. We recorded a V-band CCD time-series on 20 May, 2026 to test whether a 111.12 min baseline could recover the known positive-superhump signal and estimate an independent photometric period. We calibrated 140 images with bias, dark and normalized flat-field frames, rejected 2 artifact frames and extracted a relative-flux light curve with differential aperture photometry using 6 comparison stars from the AAVSO sequence. A single-sinusoid search detects 524.35 ± 3.3 s, the first harmonic of the published double-humped 1051.2 s superhump waveform, whereas the free double-wave Fourier model gives the strongest independent period estimate at $P = 1050.05 \pm 6.7$ s with Monte Carlo trials using empirically scaled errors. We found that the fixed 1051.2 s superhump-plus-harmonic template has the lowest effective Bayesian Information Criterion (BIC) conditional on the literature period, but the double-wave model remains the strongest independent period estimate. The data thus recover AM CVn’s known superhump and harmonic structure, but they do not tighten existing period constraints in the literature.

1 Introduction

AM CVn stars are ultracompact, hydrogen-deficient interacting binaries in which a white dwarf accretes helium-rich matter from a hydrogen-poor donor via an accretion disk. They are short-period interacting binaries with orbital periods ranging from ~ 5 –65 min. Since their angular-momentum evolution is influenced by gravitational radiation, they are studied for compact-binary evolution, helium accretion disks and low-frequency gravitational-wave astrophysics [10, 5, 1]. Accretion disks at short periods are hot and thermally stable, whereas AM CVn stars with intermediate periods can exhibit outbursts and superoutbursts [1].

AM CVn is the class prototype with a blue, hydrogen-deficient optical spectrum and optical variability on the time-scale of minutes. As such, it is widely studied for its short-period, thermally stable helium-disk variability. Early photometry in literature established a low-amplitude, double-humped ~ 17.5 min waveform with a strong ~ 8.76 min first harmonic [4, 8]. Since AM CVn’s waveforms vary between cycles, they risk appearing single-humped with phase jitter and ambiguous period estimates [8, 2].

Given this variability, existing literature distinguishes between three closely spaced clocks. Long-baseline photometry shows a dominant positive superhump at $P_{\text{sh}} \simeq 1051.2$ s, with the majority of its Fourier power in the first harmonic near 525.6 s, a negative superhump near 1011.4 s and a signal at $P_{\text{orb}} = 1028.7322$ s [7]. In contrast to the 1051 s photometric period, time-resolved spectroscopy identified the 1028.73 s signal as the true orbital period where the S-wave appears [3]. High-speed

spectroscopy similarly supports the 1051.2 s photometric modulation as a positive superhump [6]. This period structure is combined with Gaia-based distance information to revise system parameters and gravitational-wave period evolution [9].

The goals of this paper are to examine whether a short-baseline 111.12 min V-band CCD light curve of AM CVn could (1) recover the known positive-superhump and harmonic structure and (2) tighten existing period constraints in the literature. We distinguish between *independent* and *conditional* constraints, where the former fit the period freely from our data and the latter assume a literature-motivated physical identification to test for consistency.

2 Observations

We observed AM CVn (J2000 RA 12:34:54.62 and Dec +37:37:44.12) from the Stanford Student Observatory on 20 May, 2026 using the 0.7 m telescope and V-band filter. The telescope was operated with 4×4 binning and a $27' \times 18'$ field of view. The conditions were clear throughout the observations with a seeing of approximately $5''$ - $8''$. AM CVn decreased in altitude from 85° to 60° throughout the observation.

The observing sequence contained 140 CCD frames with 35, 40 and 45 s exposure times imaged from 22:20 to 00:30 PDT. The exposure times were adjusted throughout the observation night to increase signal without saturating the target and comparison stars. We excluded the last two frames as they displayed severe artifacts and low counts. The final data set thus contains 138 images and captures approximately 6.34 cycles of a 1051.2 s superhump over 111.12 min.

3 Data Reduction

We constructed a master bias frame with 10 averaged biases, and grouped 29 darks by their exposure times to form corresponding master dark frames by subtracting the master bias from averaged darks across different groups. Our flats on the observation night were contaminated with dome light, so we replaced them with 10 0.3 s V-band flats from 16 April, 2026. The borrowed flats were normalized and had raw median levels of approximately 2.05×10^4 to 2.52×10^4 ADU with 99th-percentile levels below saturation. Since the flats were taken on a different night, the residual flat-field error is a potential source of systematic error. We used master dark frames with exposure times of 0.3 s, 35 s, 40 s and 45 s to calibrate the flat and science frames with

$$I_{\text{cal}} = \frac{I_{\text{raw}} - B - D(t_{\text{exp}})}{F}.$$

We previously excluded the last two frames as they displayed severe artifacts and low counts during the observation night, and plate-solved the remaining 138 V-band frames with ASTAP. We relied on the AAVSO comparison sequence X42550AJH to define the sky coordinates of AM CVn and its six comparison stars with magnitudes lower than 15 to have higher signal to noise ratios. We then used the WCS to predict their pixel positions and refined them by local centroiding within an 8 pixel search radius. AM CVn has a median WCS-to-centroid offset of 1.20 pixels, whereas the comparison stars have median offsets of 0.52 to 5.47 pixels.

Aperture photometry was performed on the target and comparison stars using a 6 pixel source aperture and 12-20 pixel sky annuli. We estimated local sky levels from the annuli and subtracted them from the aperture sums before dividing the background-subtracted fluxes by exposure times to convert them to count rates. In order to remove seeing variations across different exposures, we constructed an ensemble differential light curve by normalizing each star to its median count rate.

We then computed the comparison ensemble by averaging the normalized comparison star fluxes and calculated the AM CVn relative flux by dividing the target count rate by the comparison ensemble.

We validated the comparison stars with a leave-one-out comparison star control light curve, where we treated each comparison star as a pseudo-target and divided it by an ensemble composed of the remaining comparison stars (Figure 1). Comparison stars 90, 96, 125, 143 and 144 had robust scatter $\sim 1\%$, whereas comparison star 112 had a scatter of $\sim 2.9\%$. Since comparison star 112 was the noisiest control, we repeated the differential photometry and period fit excluding it. The free double-wave period changed by 0.40s, which is smaller than the Monte Carlo period uncertainty of 6.7s, so we retained the full six-star ensemble for the primary reduction. These controls show that differential photometry did not introduce AM CVn-like variability into the comparison stars.

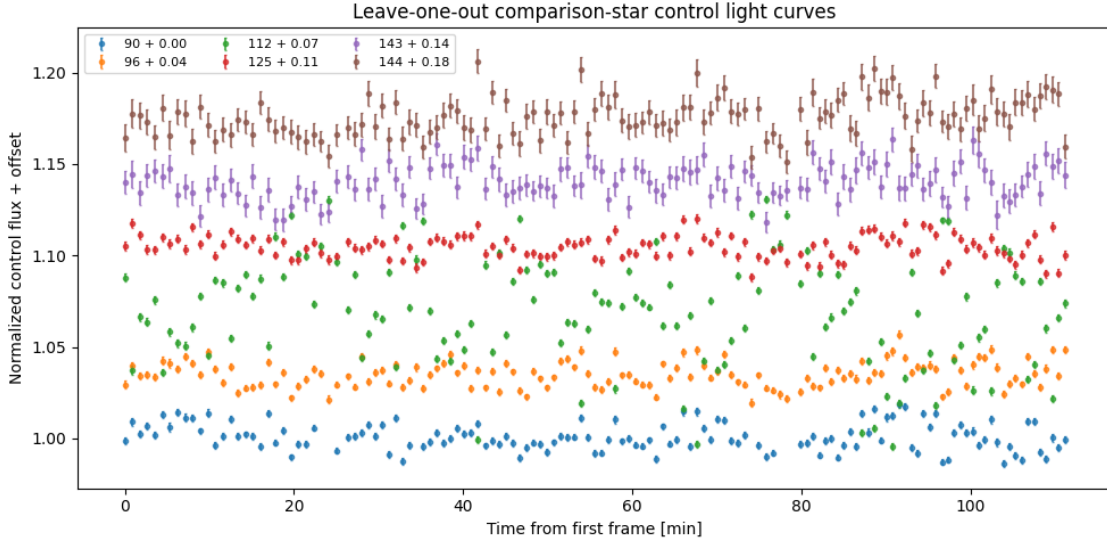


Figure 1: Leave-one-out comparison-star control light curves offset vertically for clarity. Each comparison star is treated as a pseudo-target and divided by the remaining ensemble. Stars 90, 96, 125, 143 and 144 show robust scatter of $\sim 0.6 - 1.1\%$, while star 112 shows 2.9% consistent with its larger centroiding offset. None of the controls shows AM CVn-like variability, indicating that the differential-photometry procedure does not introduce the detected variability.

The AM CVn differential light curve contains 138 points over (111.12 min), corresponding to 6.34 cycles of a (1051.2s) superhump. The robust scatter of our target light curve is 1.64%, whereas the median formal photometric error is 0.55%. The larger observed scatter indicates that there may be residual atmospheric, flat-field, tracking and comparison-ensemble systematic errors that contributed to the uncertainty. We thus used empirical error scaling when analyzing the period.

4 Analysis Methods

We analyzed the differential light curve with a ladder of models, using constant relative flux as our baseline model. The periodic models are a free single-sinusoid model and a free double-wave Fourier model with the fundamental and its first harmonic. We used Lomb-Scargle periodograms as diagnostic period search tools [11], matching the two-term periodogram to AM CVn’s superhump-dominated light curve.

The literature-conditioned models relied on published AM CVn’s 1028.7322s orbital period,

1051.2 s positive-superhump period and 1011.4 s negative-superhump period. We also fit a fixed superhump-plus-harmonic template at 1051.2 s and a beat-constrained superhump proxy in which $f_{\text{sh}} = f_{\text{orb}} - f_{\text{prec}}$. Although this is in line with the standard positive-superhump interpretation, it is a conditional physical prior rather than a direct measurement of the disc precession period. The orbital and superhump periods in literature gives an expected apsidal precession period of $\simeq 13.37$ h, which means that our observation only covers 0.14 cycle. As such, this fitted precession parameter is a beat-relation proxy.

For a selected period, our fitted light-curve is a weighted Fourier model given by the equation

$$m(t_i) = c + \sum_{j=1}^M [a_j \sin(2\pi h_j t_i / P_j) + b_j \cos(2\pi h_j t_i / P_j)],$$

where t_i is time from first exposure and y_i is normalized differential flux. The coefficients minimize

$$\chi^2 = \sum_{i=1}^N \left[\frac{y_i - m(t_i)}{\sigma_i} \right]^2.$$

Although both the fixed 1051.2 s and free double-wave models rely on this mathematical form, the template uses a literature period, whereas the free model fits a period from our data.

We compared the models with an effective Bayesian information criterion (BIC),

$$\text{BIC}_{\text{eff}} = \chi^2 + k_{\text{eff}} \ln N, \quad k_{\text{eff}} = k_{\text{lin}} + k_{\text{nonlin}},$$

where $k_{\text{lin}} = 1 + 2M$ counts the constant, sine and cosine coefficients and k_{nonlin} counts either the free or precession period. The $\Delta\text{BIC}_{\text{eff}}$ in Table 1 is measured relative to the free double-wave model, such that negative values correspond to lower effective BIC. That being said, it is a heuristic because our period search is nonlinear. We also computed a time-blocked cross-validation error with contiguous time blocks to examine whether increased model flexibility improved predictions.

We computed χ^2 intervals for the free periods and performed a parametric Monte Carlo period-uncertainty test. In each Monte Carlo trial, we preserved the original observing times and generated a synthetic light curve from the best-fit double-wave model using scaled photometric errors, rerunning the Lomb-Scargle and free-period model searches. Since our observed scatter exceeds the formal photometric errors (reduced $\chi^2 \simeq 5.2$ for the best model), we rescaled the formal errors by $\sqrt{\chi^2_{\nu}} = 2.27$ before calculating period uncertainties. We also tested the stability by fitting the two halves separately, clipping $|z| > 4$ outliers, dropping comparison stars, adding nuisance regressors, using robust loss functions and examining the resulting periodogram.

5 Results

Our differential light curve contains 138 points over 111.12 min, which corresponds to 6.34 cycles of a 1051.2 s superhump. The free single-sinusoid search recovers

$$P_{\text{sine}} = 524.35 \pm 3.3 \text{ s}$$

with uncertainty from Monte Carlo trials. The single-term Lomb-Scargle periodogram detects the same harmonic at 524.97 s with a false-alarm probability of 1.8×10^{-5} (Figure 2), whereas the two-term periodogram recovers a fundamental near 1053 s. A single sinusoid is thus sufficient to detect AM CVn, but a multi-harmonic or physically motivated model is necessary to recover the underlying double-wave period.

The best independent period measurement is the free double-wave fit,

$$P_{\text{free}} = 1050.05 \pm 6.7 \text{ s},$$

which is consistent with the published 1051.2 s positive-superhump period within the Monte Carlo 1σ uncertainty. That being said, it is not yet precise enough to improve the published photometric or spectroscopic periods in literature.

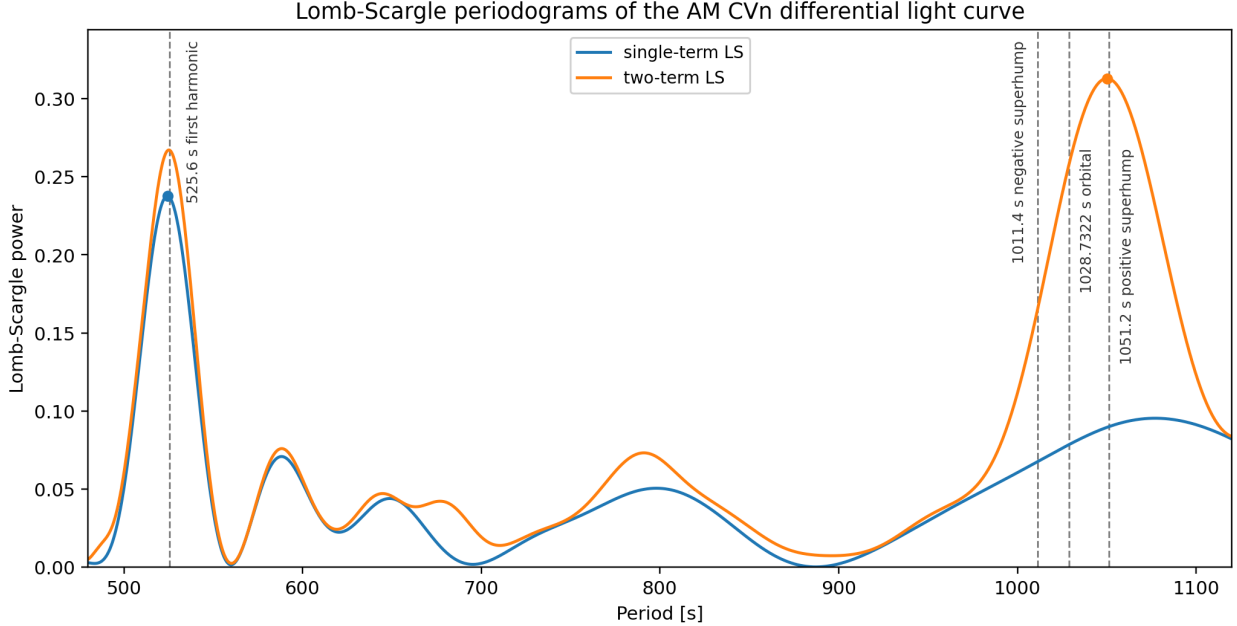


Figure 2: Single-term (blue) and two-term (orange) Lomb-Scargle periodograms of the AM CVn differential light curve. Dashed lines indicate the published 525.6s first harmonic, 1011.4s negative superhump, 1028.7322s orbital period and 1051.2s positive superhump. The dominant single-term peak near 525s is the first harmonic of the double-humped superhump waveform, whereas the two-term periodogram recovers the fundamental as a single peak near 1053s.

Table 1: Model ladder for AM CVn. $\Delta\text{BIC}_{\text{eff}}$ is measured relative to the free double-wave model, where negative values indicate lower effective BIC. The fixed 1051.2s model gives the lowest effective BIC, whereas the free double-wave row is the best independent period measurement.

Model	Period treatment	Period (s)	k_{eff}	RMS (%)	BIC_{eff}	$\Delta\text{BIC}_{\text{eff}}$
Constant flux	none	—	1	1.50	1006.40	288.75
Free single sine	fitted harmonic	524.35	4	1.31	783.12	65.47
1051.2 s + harmonic	fixed literature superhump	1051.2	5	1.25	712.91	−4.74
Beat-constrained superhump proxy	fitted beat proxy	1050.03	6	1.25	717.65	0.00
Free double wave	fitted fundamental + harmonic	1050.05	6	1.25	717.65	0.00

Although the fixed 1051.2s superhump-plus-harmonic model has the lowest effective BIC, it is conditioned by the literature period. As such, the free double-wave model is the strongest independent measurement. Its relatively higher effective BIC thus reflects the additional period-search degree of freedom. The beat-constrained superhump proxy has a fitted photometric period

of 1050.03 s, which corresponds to a beat-relation precession proxy of $\simeq 14.09$ h. However, it differs from the published $\simeq 13.37$ h, since our baseline only covers 0.14 precession cycle. This model illustrates how a beat period is sensitive to second-scale shifts in the photometric period.

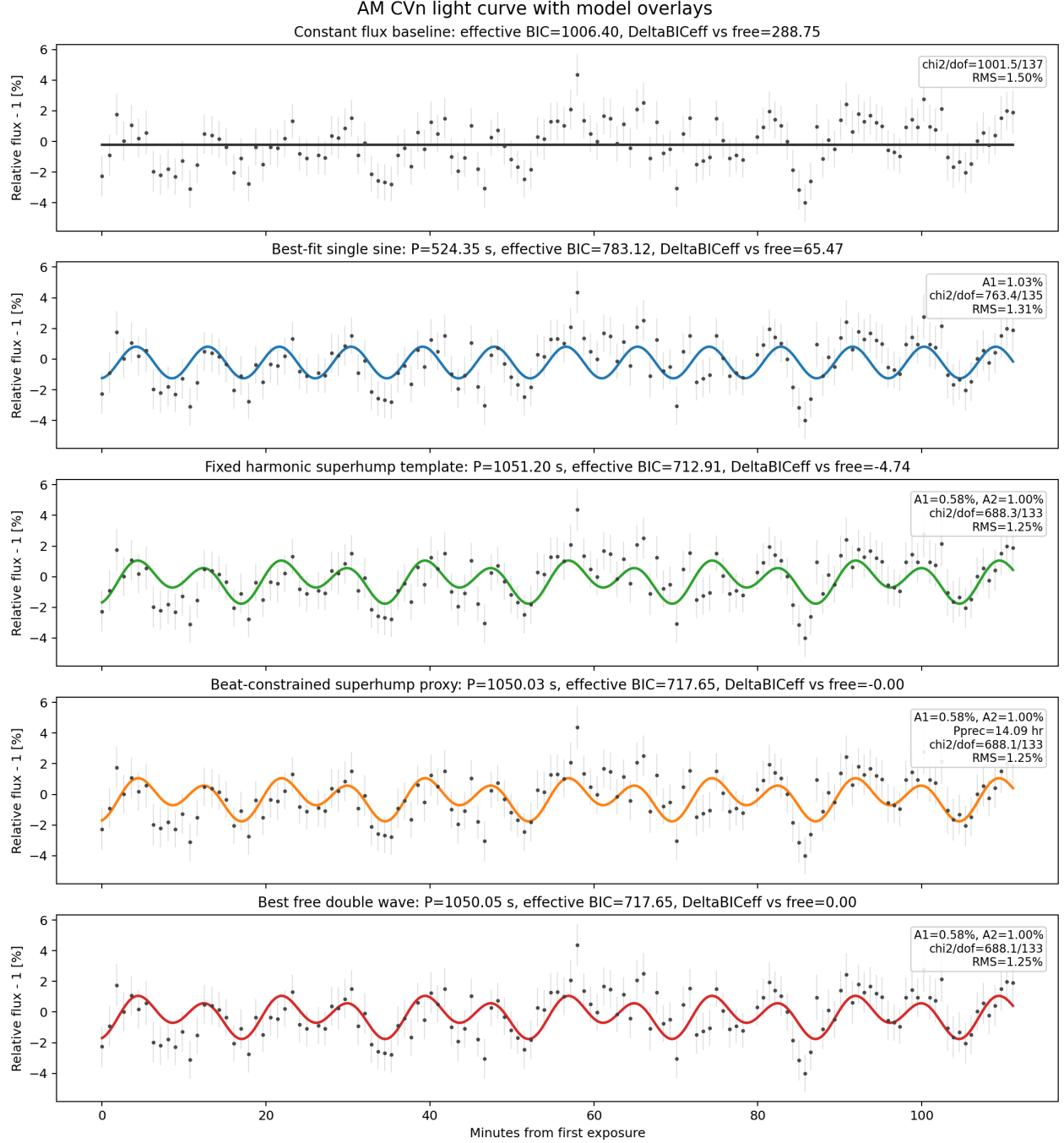


Figure 3: Relative V-band light curve (grey points with formal error bars) with the five fitted models overplotted: constant flux, best single sinusoid (524.35s), fixed 1051.2s superhump-plus-harmonic, beat-constrained superhump proxy (1050.03s) and free double wave (1050.05s).

The period result is robust to photometric changes, but it is limited by the short baseline. Clipping $|z| > 4$ outliers changes the double-wave period by 0.08 s, and excluding the noisiest comparison star 112 shifts it by 0.40 s. That being said, these shifts are comfortably within the 6.7 s Monte Carlo uncertainty. Similarly, the residual periodogram has its strongest peak near 35 min with false-alarm probability 0.10, indicating that no statistically significant unmodelled signal remains. However, the first- and second-half fitted periods differ by 16.4 s, which suggests that this light curve does not tightly constrain the period by itself.

Additional model complexity did not affect the physical result, but nuisance regressors for time, image shifts, exposure state, comparison count and common mode reduced the RMS from 1.25% to 1.10%, lowered the BIC and improved blocked cross-validation error. Since the recovered period stayed within ~ 1048 – 1056 s throughout, we treat these regressors as systematic diagnostics as opposed to a stronger physical model. Similarly, robust soft- L_1 , Huber and Cauchy losses shifted the period to 1050.03–1050.34 s, and comparison star subsets decreased the RMS to 1.09% with periods spanning ~ 1048 – 1056 s. These tests suggest that the limiting uncertainty remains the short baseline and intrinsic AM CVn waveform variability as opposed to Gaussian photon noise. Since the independent period uncertainty at 6.7 s is larger than the published precision of AM CVn clocks, our light curve independently recovers the known positive-superhump and first-harmonic structure without tightening existing period constraints.

6 Discussion and Limitations

We anchored our model comparison to published data for AM CVn, where the periods calculated from the models are consistent. The single-sinusoid search and single-term Lomb-Scargle periodogram detect the ~ 524 s harmonic with a false-alarm probability 1.8×10^{-5} , whereas the free double-wave model recovers 1050.05 ± 6.7 s within 1σ of the published 1051.2 s superhump and the literature-conditioned superhump-plus-harmonic model gives the lowest effective BIC. Similarly, the fitted waveform is morphologically consistent with the literature with the first-harmonic amplitude (1.00%) exceeding the fundamental (0.58%), where most of the Fourier power for the double-humped waveform is in the first harmonic [7].

That being said, our main limitation is the short baseline as 111.12 min of observation was only able to capture 6.34 cycles and 0.14 of the expected disc-precession cycle. This limits the frequency resolution of $\delta P \simeq P^2/T \approx 165$ s at $P \simeq 1050$ s for $T = 111.12$ min, which is four to ten times larger than the 17 s to 40 s separations between the published negative superhump, orbital and positive superhump. As such, our data are not able to distinguish between them; this is illustrated in Figure 2, where the three features are visualized as a single peak. Our recovery of the ~ 1050 s signal as the positive superhump and the $\simeq 14.1$ h value from our beat-constrained proxy are thus literature-conditioned interpretations. Separating the superhump from the orbital period photometrically would require a baseline of at least one disc-precession cycle of $\simeq 13.4$ h.

The error tests in Section 5 localize the limiting uncertainties, indicating that the largest instability is the 16.4 s difference between the first- and second-half fitted periods. This is approximately $2.45\times$ the Monte Carlo uncertainty consistent with the cycle-to-cycle waveform changes in AM CVn [8, 2]. As such, we do not claim a tight independent constraint. Similarly, there are also residual low-frequency systematic errors from airmass, tracking and flat-field errors from borrowed flats, evidenced by how nuisance regressors for time, tracking and exposure state lower the RMS, BIC and improve the blocked cross-validation error.

Another limitation is that broadband photometry is not able to detect AM CVn’s S-wave emission directly, as it is a spectroscopic velocity feature associated with the stream-disc impact

region. As such, testing it requires time-resolved spectroscopy or Doppler tomography. Our V-band photometry was thus able to obtain data consistent with the accreting white dwarf interpretation but not able to independently establish orbital clocks or classifications. Future studies could thus focus on increasing the baseline such that it spans at least one $\simeq 13.4$ h precession cycle to resolve the superhump-orbital triplet. Another direction would be to complement V-band imaging with time-resolved spectroscopy to detect AM CVn’s S-wave emission directly.

7 Conclusions

Our 111.12 min V-band light curve successfully detects short-timescale AM CVn variability, where the single-sinusoid model detects the ~ 524 s first harmonic of the published double-humped waveform. The best independent fit gives $P_{\text{free}} = 1050.05 \pm 6.7$ s, whereas the literature-conditioned 1051.2 s superhump-plus-harmonic model gives the best conditional description with the lowest effective BIC. Both the free fit and beat-constrained superhump proxy have the same BIC.

Although our data do not tighten existing period constraints in literature, analyzing the data shows that the main limiting factor is the short baseline and intrinsic AM CVn waveform variability as opposed to photometric noise. Our short-baseline photometry thus independently recovers the variability consistent with AM CVn’s published superhump and harmonic structure.

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A Individual Contributions

We collaborated closely on all parts of the project. In analyzing the data, we each wrote code to analyze the data independently and compared our results to improve the combined final code. We also read literature and shared our findings to interpret and write the report. We used AI to debug our code, refine report writing and clarify concepts from lectures and literature.